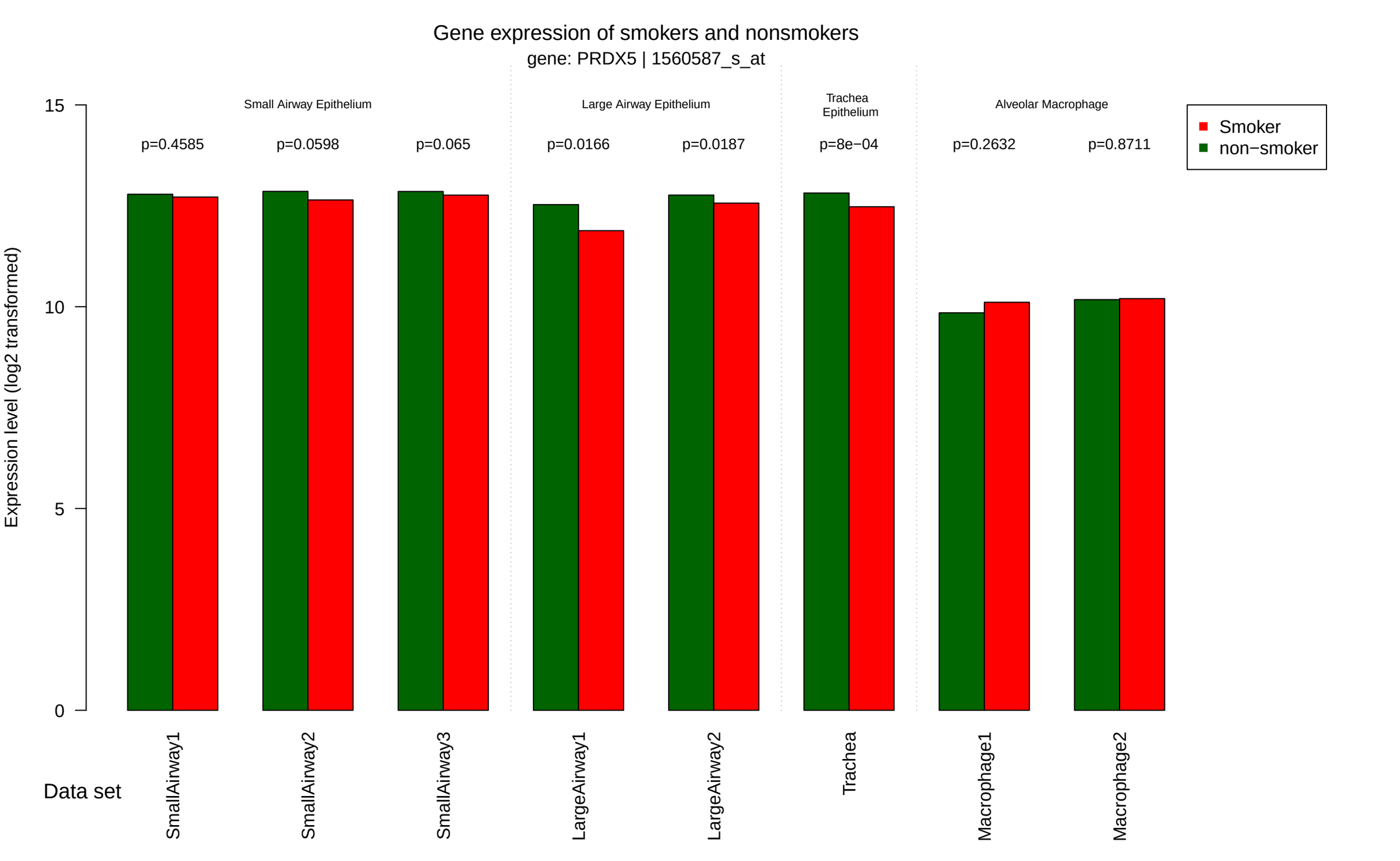
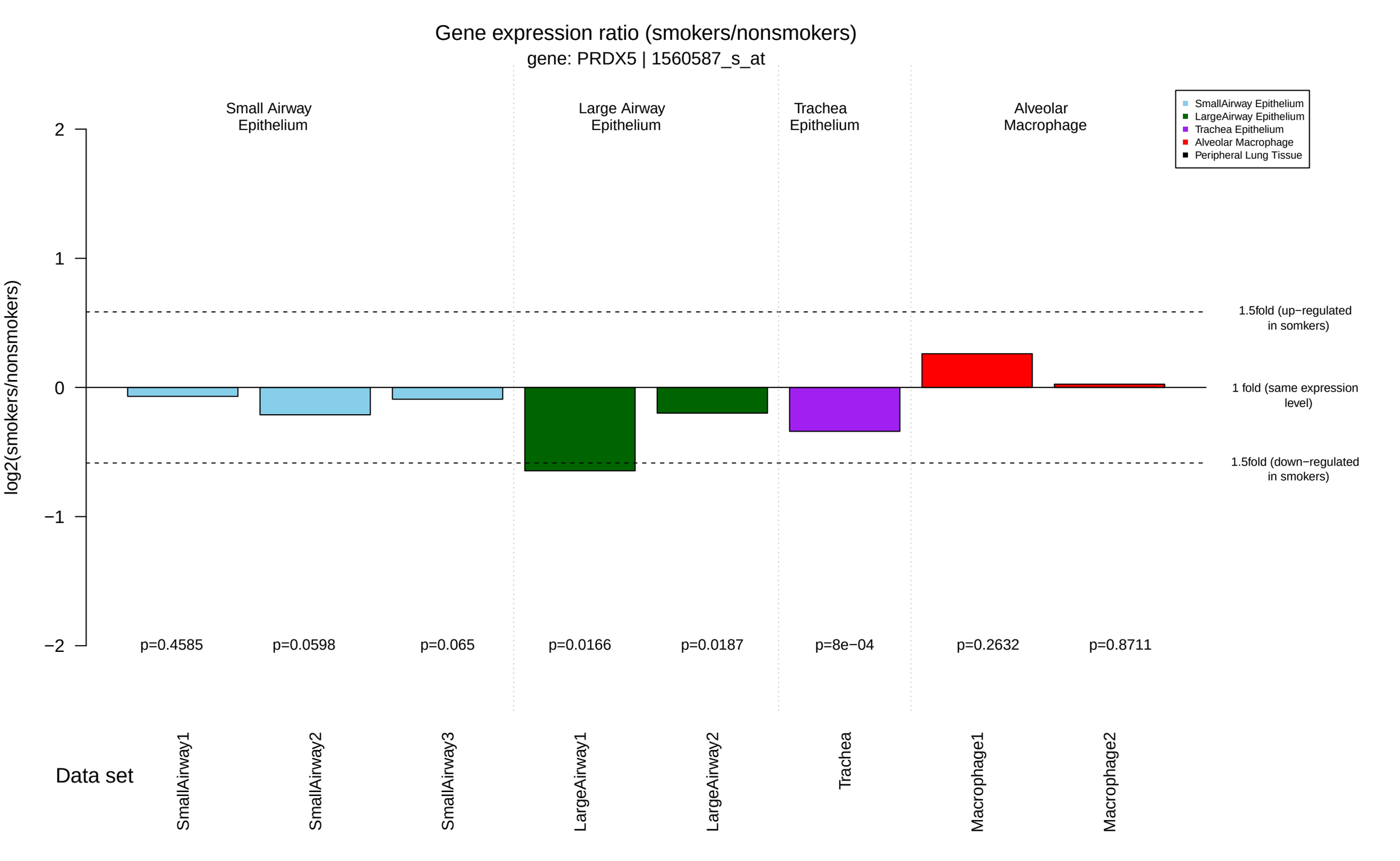
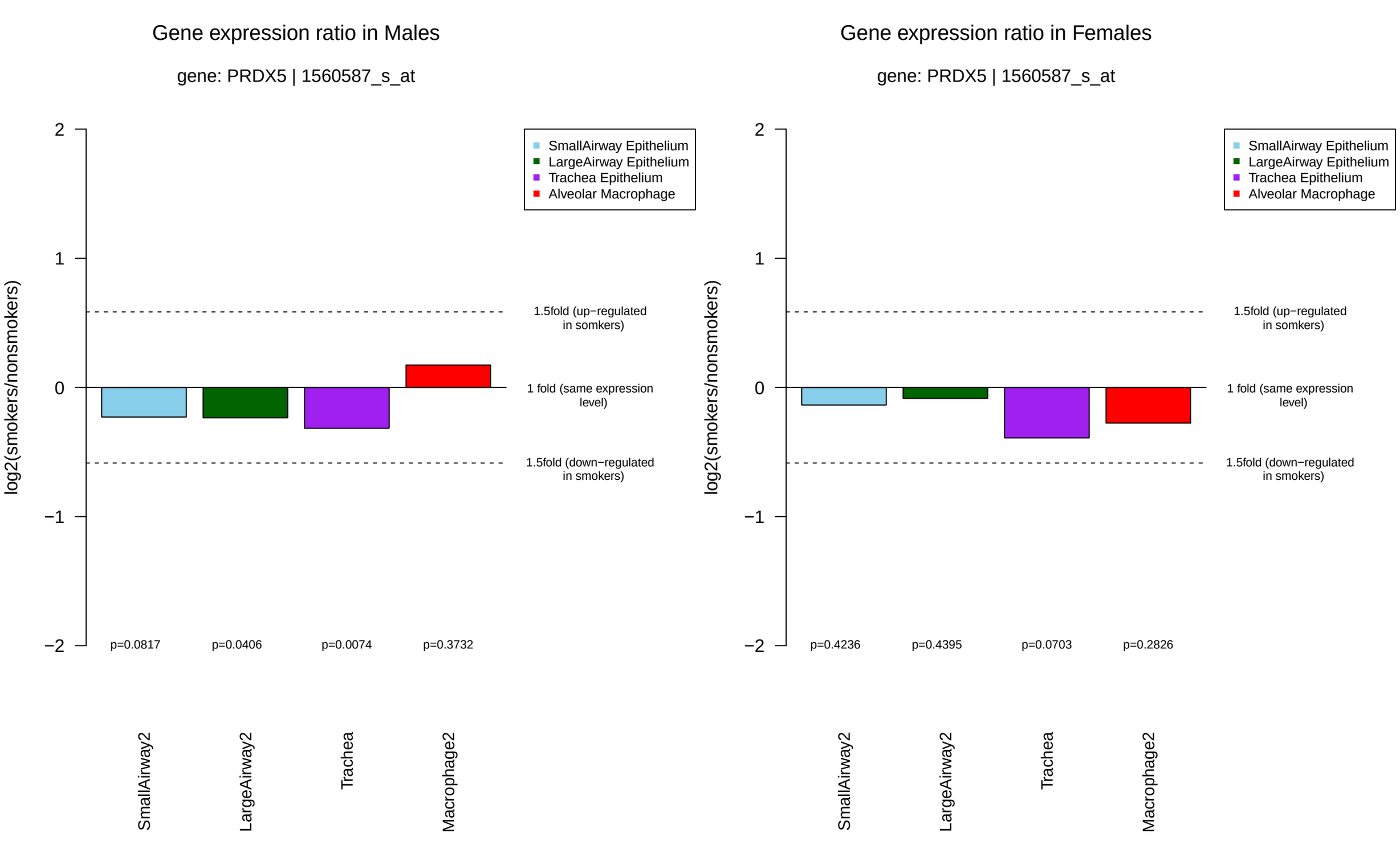
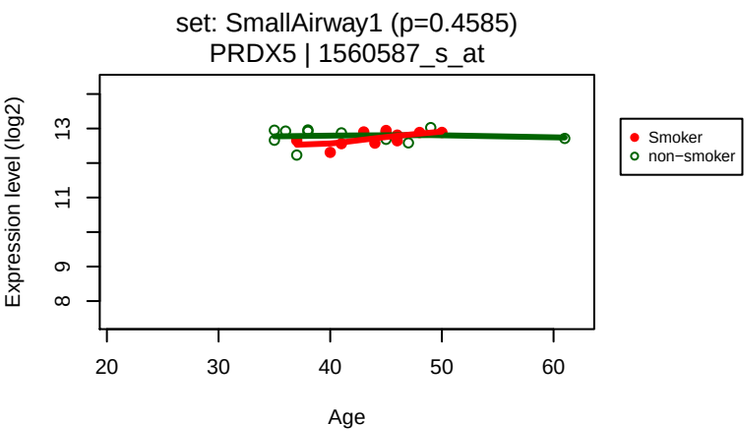
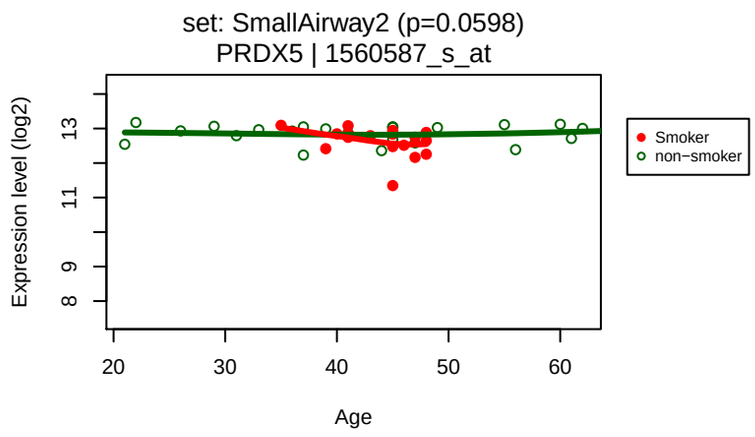
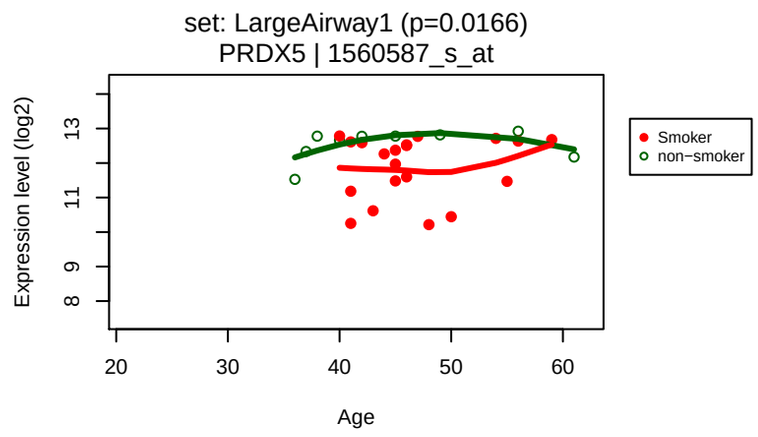

Gene expression of smokers and nonsmokers
gene: PRDX5 | 1560587_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4585
p=0.0598
p=0.065
p=0.0166
p=0.0187
p=8e−04
p=0.2632
p=0.8711
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

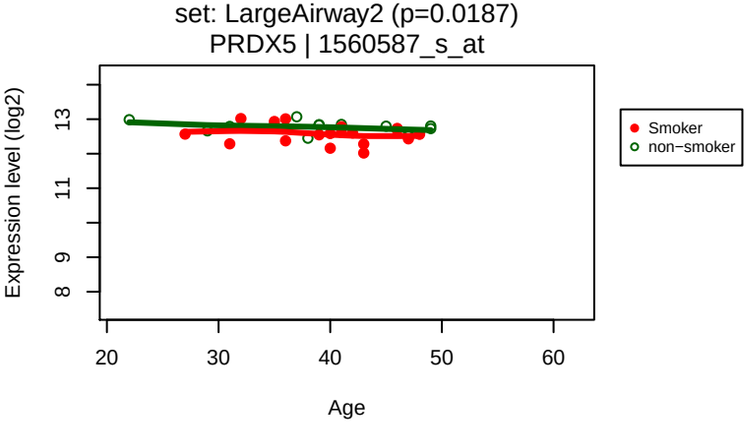

Gene expression ratio (smokers/nonsmokers)
gene: PRDX5 | 1560587_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4585
p=0.0598
p=0.065
p=0.0166
p=0.0187
p=8e−04
p=0.2632
p=0.8711
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

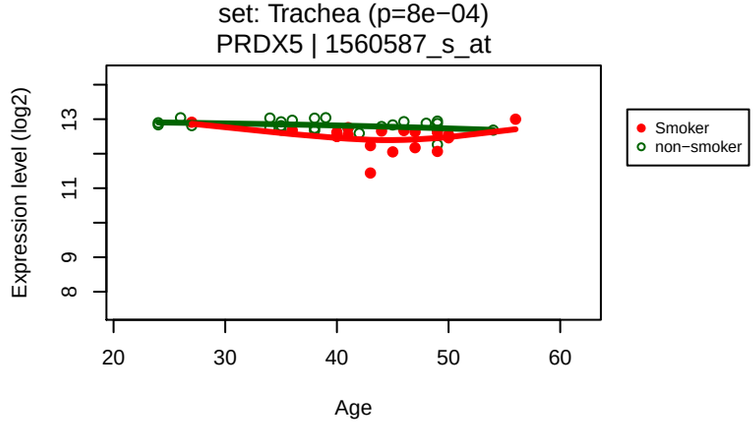

Gene expression ratio in Males
gene: PRDX5 | 1560587_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0817
p=0.0406
p=0.0074
p=0.3732
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRDX5 | 1560587_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4236
p=0.4395
p=0.0703
p=0.2826
SmallAirway2
LargeAirway2
Trachea
Macrophage2

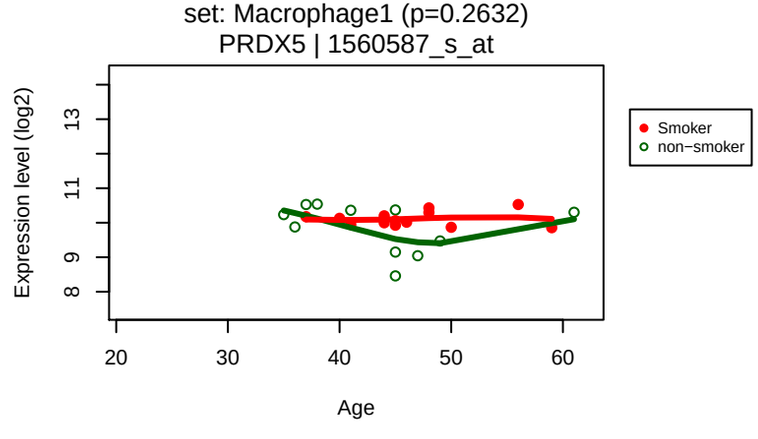

set: SmallAirway1 (p=0.4585)
PRDX5 | 1560587_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

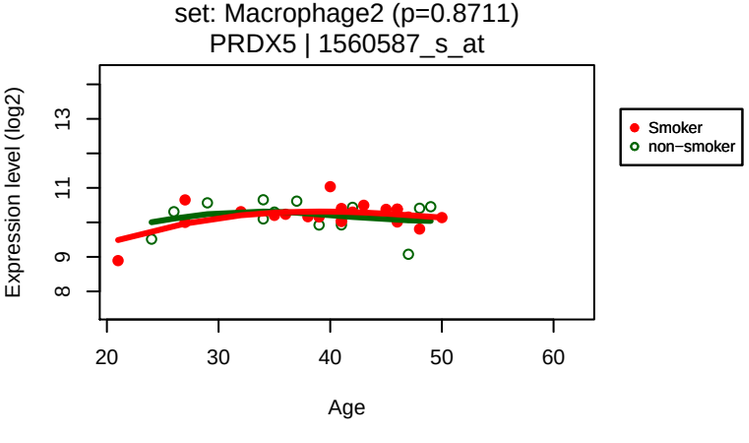

set: SmallAirway2 (p=0.0598)
PRDX5 | 1560587_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

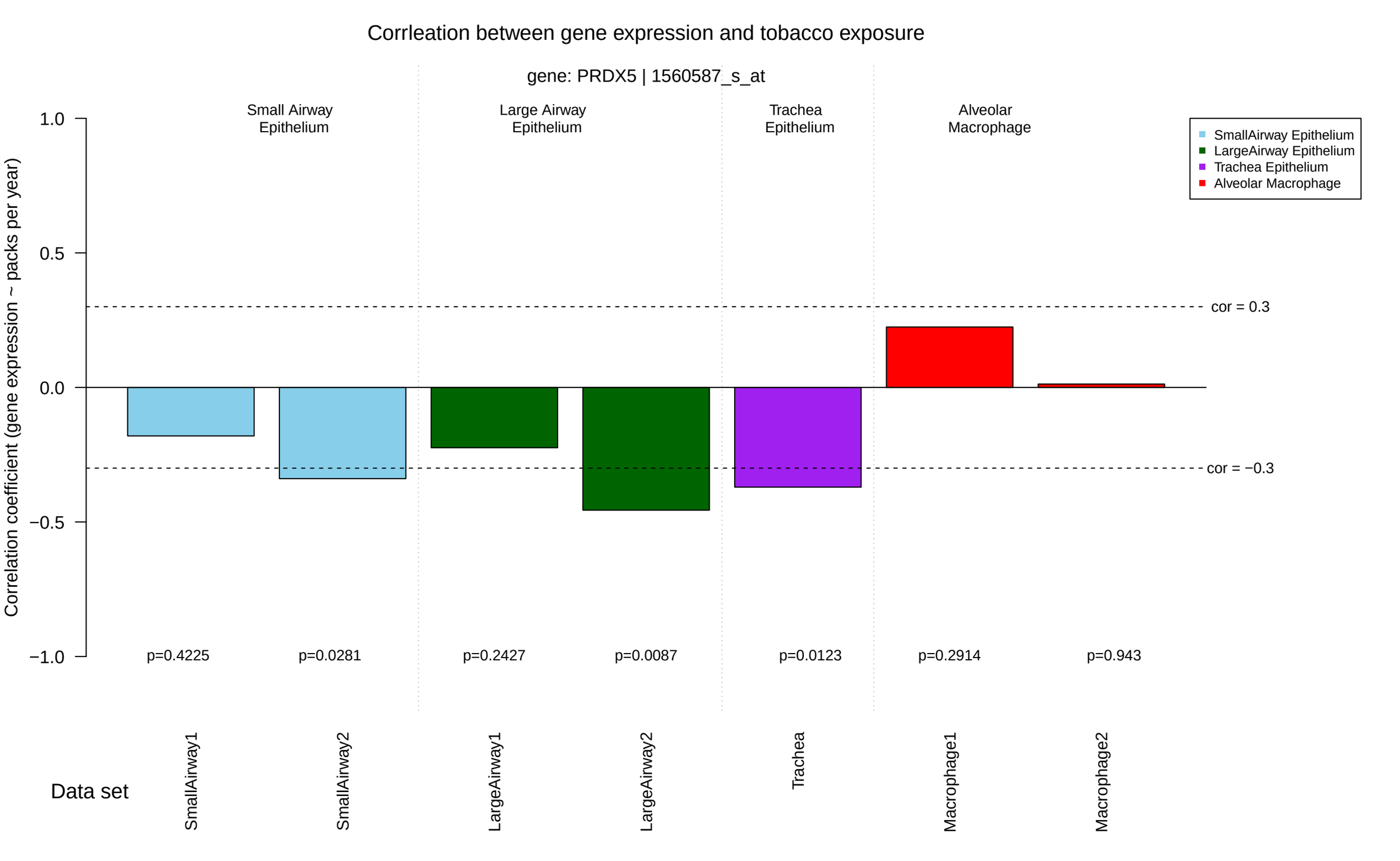

set: LargeAirway1 (p=0.0166)
PRDX5 | 1560587_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0187)
PRDX5 | 1560587_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=8e−04)
PRDX5 | 1560587_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2632)
PRDX5 | 1560587_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8711)
PRDX5 | 1560587_s_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRDX5 | 1560587_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4225
p=0.0281
p=0.2427
p=0.0087
p=0.0123
p=0.2914
p=0.943
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2